

Deep Learning Project Proposal

BTC/USDT 1-hour Forecasting

Yu ByeongHyeon (20211199)

CSE403: Deep Learning, UNIST

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- **Why Bitcoin (BTC/USDT) over stocks?**

- **Data access:** Reliable and free via Binance Public API.
- **Structure:** Mostly numerical time-series (OHLCV), simpler preprocessing than stocks that often require textual/fundamental data.
- **24/7 market:** Continuous signal without market-close gaps.

- **Research Motivation**

- The motivation is to test whether deep learning can capture patterns in such a noisy and volatile dataset, without relying on external data like news.

Problem Definition

- **Task:** Given past 168 hours (1 week) of features, predict the **next 1-hour log return**.
- **Research Question:** Do DL models (GRU/LSTM) outperform naive baselines for short-horizon forecasting?
- **Why it matters:**
 - Crypto exhibits regime shifts and volatility clustering.
 - Evaluates whether DL extracts predictive structure from price/volume only.

Dataset (Overview)

- **Source:** Binance Public API, BTC/USDT, 1-hour candles (OHLCV).
- **Period:** Oct 15, 2023 – Oct 15, 2025 (**2 years**).
- **Size:** $\sim 17,520$ rows ($24 \times 365 \times 2$).
- **Scope:** Use 2 years first; **plan to extend to 3 years** to examine longer-horizon effects on stability/generalization.

Dataset Summary

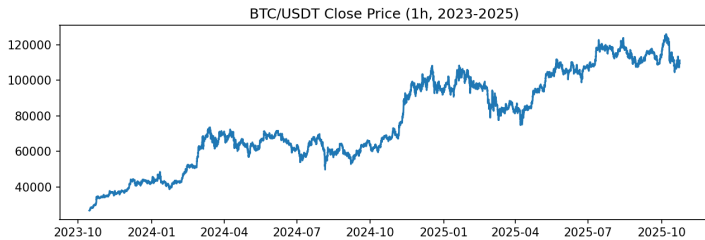
Source	Binance Public API, BTC/USDT, 1-hour candles
Period	Oct 15, 2023 – Oct 15, 2025 (2 years)
Size	~17,520 rows ($24 \times 365 \times 2$)
Features	• OHLCV: Open, High, Low, Close, Volume • Log returns: $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$ • Derived (planned): EMA, RSI, MACD, ATR, time-of-day, day-of-week
Label	Next 1-hour log return (r_{t+1})

Dataset Example (CSV snapshot)

1	timestamp	open	high	low	close	volume	time	log_return
2	1.69733E+12	26852.48	26853.45	26815.76	26848.76	255.5488	2023-10-15 0:00	
3	1.69733E+12	26848.76	26916.77	26845.4	26890.01	511.8841	2023-10-15 1:00	0.001535
4	1.69734E+12	26890	26904.32	26871.65	26883.6	235.7798	2023-10-15 2:00	-0.00024
5	1.69734E+12	26883.6	26907.16	26877.17	26880.46	319.3487	2023-10-15 3:00	-0.00012
6	1.69734E+12	26880.46	26889.22	26871.86	26883.98	239.7693	2023-10-15 4:00	0.000131
7	1.69735E+12	26883.98	26886.08	26881.26	26882.44	129.759	2023-10-15 5:00	-5.73E-05
8	1.69735E+12	26882.45	26882.45	26873.94	26879.53	198.9666	2023-10-15 6:00	-0.00011
9	1.69735E+12	26879.53	26882.52	26873.91	26882.52	188.6514	2023-10-15 7:00	0.000111
10	1.69736E+12	26882.51	26910	26882.51	26905.79	368.3912	2023-10-15 8:00	0.000865
11	1.69736E+12	26905.78	26919	26897.56	26901.29	264.7904	2023-10-15 9:00	-0.00017
12	1.69736E+12	26901.3	26901.3	26893	26895	279.829	2023-10-15 10:00	-0.00023
13	1.69737E+12	26895	26895.01	26843.79	26852.47	374.8463	2023-10-15 11:00	-0.00158
14	1.69737E+12	26852.48	26852.48	26826.7	26838.14	368.61	2023-10-15 12:00	-0.00053
15	1.69737E+12	26838.14	26965.28	26808.25	26861.13	1432.35	2023-10-15 13:00	0.000856
16	1.69738E+12	26861.12	26918.62	26857.15	26873.37	457.0334	2023-10-15 14:00	0.000456
17	1.69738E+12	26873.38	26906.89	26851.98	26898.87	416.1936	2023-10-15 15:00	0.000948
18	1.69739E+12	26898.87	26934.58	26893.41	26930.29	383.2417	2023-10-15 16:00	0.001167
19	1.69739E+12	26930.29	27017.44	26920.55	26982.49	1043.593	2023-10-15 17:00	0.001936
20	1.69739E+12	26982.49	27063.08	26956.17	27056.41	682.5503	2023-10-15 18:00	0.002736
21	1.6974E+12	27056.4	27081	26986.07	27043.75	668.3181	2023-10-15 19:00	-0.00047
22	1.6974E+12	27043.75	27234.99	27015	27206.21	2355.661	2023-10-15 20:00	0.005989

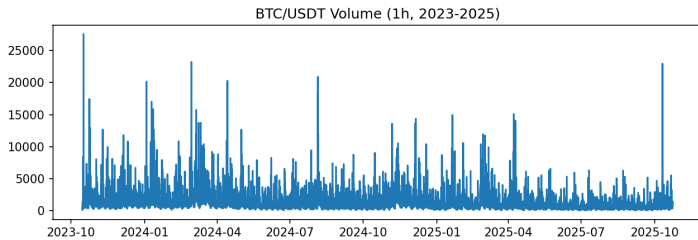


Exploratory Data Analysis: Close



figureBTC/USD Close (1h, Oct 2023–Oct 2025)

Exploratory Data Analysis: Volume



figureBTC/USDT Volume (1h, Oct 2023–Oct 2025)

Preprocessing Plan

- **Cleaning:** UTC-aligned, sorted, deduplicated; missing values: negligible (forward-fill or drop if any).
- **Core feature:** Log return $r_t = \log\left(\frac{C_t}{C_{t-1}}\right)$; consider HL range, technical indicators (no leakage).
We use log returns instead of raw prices because they are additive over time, symmetric, and generally more stable for modeling
- **Scaling:** Fit scaler on *train* only; transform val/test.
- **Windowing:** Sliding window of 168 steps (1 week) $\Rightarrow$ target r_{t+1} .

Chronological Split (No Leakage)

Split	Date Range (UTC)	Purpose
Train	Oct 15, 2023 – May 31, 2025	Model training, scaler fitting
Validation	Jun 1, 2025 – Aug 31, 2025	Hyperparam tuning, early stopping
Test	Sep 1, 2025 – Oct 15, 2025	Final unbiased evaluation

Note: Time order preserved; no random shuffling. Indicators/scalers fitted on train only; applied to val/test.

Baselines and Models

- **Baselines:** Last value (naive), SMA/EMA.
- **DL candidates:** GRU, LSTM (*primary*); TCN / lightweight Transformer (*optional*).
- **Input shape:** (batch, 168, features) **Target:** r_{t+1} .

- **Regression:** MAE, RMSE, MAPE (if needed).
- **Directionality:** Up/Down accuracy, F1 (with threshold on r_{t+1}).
- **Diagnostics:** Error by volatility regime, by volume spikes.

Risks & Mitigations

- **Regime shifts:** Past patterns may not transfer to recent periods.
Mitigation: Recent-weighted training, rolling validation, add regime-aware features.
- **Data size (2y):** Limited for large Transformers.
Mitigation: Start with GRU/LSTM; **plan to extend to 3y** and/or add multi-asset data.
- **Overfitting:** High flexibility vs. noisy labels.
Mitigation: Early stopping, dropout, small models, proper scaling, leakage checks.

Thank you!